CAR HIRE MANAGEMENT SYSTEM

Proposed Solution Design

Version 1.0

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Prepared for	Doxa Car Hire Services
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1. Project Overview

This document outlines the proposed technology stack, functional requirements, non-functional requirements, and implementation approach for the development of a modern Car Hire Management System.

The proposed system is designed to digitize and automate key car hire operations, including client management, vehicle dispatch, booking extensions, check-ins, owner statements, notifications, and financial reporting.

2. Recommended Technology Stack

Component	Recommended Technology
Frontend	Next.js (React + TypeScript)
Backend	Node.js + Express
Database	PostgreSQL
Authentication	OTP / Secure Login
Hosting	Cloud Hosting (Vercel / VPS)
Notifications	SMS API Integration

3. Why This Stack Is Recommended

- Modern and scalable architecture
- Fast performance on desktop and mobile devices
- Secure user authentication
- Easy future expansion (mobile app, M-Pesa integration, analytics, etc.)
- Lower maintenance cost compared to older legacy systems

4. Functional Requirements (Phase 1 / MVP)

Client Management

- Register clients
- View and edit client details
- Suspend or activate accounts
- Generate client receipts

Vehicle Management

- Add vehicles
- Assign vehicles to owners
- View fleet status
- Track parked and hired vehicles

Booking & Dispatch

- Create bookings
- Dispatch vehicles
- Extend hire periods
- Check in returned vehicles
- Calculate lateness penalties

Owner Management

- Create owner accounts
- Generate owner earnings statements
- Generate owner expense statements
- View payment history

Finance & Reports

- Record income and expenses
- Generate basic profit and loss reports
- Export reports to Excel/CSV

Notifications

- Send booking confirmations
- Send return reminders
- Send owner notifications

Staff Management

- Create staff accounts
- Assign user roles
- Control access permissions

5. Non-Functional Requirements

Requirement	Description
Performance	Dashboard pages load within 3 seconds
Security	Secure login and protected user sessions
Reliability	Data stored safely in a centralized database with regular backups
Usability	Simple and user-friendly interface
Scalability	Supports future growth in vehicles and clients
Availability	Accessible from both desktop and mobile devices
Maintainability	Code structure allows future enhancements

6. Proposed Development Phases

Phase	Duration
Requirements & Planning	3–5 Days
UI/UX Design	5–7 Days
Backend Development	3 Weeks
Frontend Development	3 Weeks
Total Development	6 Weeks
Testing & Deployment	1 Week

7. Project Cost & Budget

Item	Amount (KSh)
Development Cost (Phase 1 MVP)	40,000
Initial Deposit (50%)	20,000
Balance Upon Completion	20,000
Total	**40,000**

8. Deliverables

- Responsive web-based Car Hire Management System
- Admin dashboard
- Client management module
- Vehicle management module
- Booking and dispatch module
- Basic finance reporting module

- Owner statement module
- SMS notification integration
- Deployment and basic training

9. Conclusion

The proposed solution is technically feasible, scalable, and suitable for deployment in a modern car hire business environment. The selected technology stack provides a strong foundation for future enhancements such as M-Pesa payments, mobile applications, advanced analytics, and automated accounting features.

We look forward to working with Doxa Car Hire Services to deliver a reliable and efficient car hire management platform.

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